MAM · Exponential and Logarithmic Functions — Patterns and Solution Strategies

Based on: 42 past exam questions from 2016–2025 containing exponentials/logarithms, marking keys + exam reports (the 2016, 2018, 2022–2025 reports all list logarithm-related content as a weakness: log laws 2016/2024/2025; logarithmic graphs 2018/2024; reading and interpreting logarithmic scales 2023; converting between exponential and logarithmic form 2022). Original questions are in `topics/04_Exp_Log.pdf`; also cites 2 questions from the 2016 sample paper (`2016S-A10`, `2016S-A13`, original papers are in `papers/2016Sample_*`). Review status: has undergone one round of independent review and was revised according to feedback (see `Review_log.md`).

Toolbox

- Definition: $y = \log_a x \Leftrightarrow a^y = x$. The first step in many questions is converting between logarithmic and exponential form (the key often has 1 mark for "converts to index form / uses the inverse relationship").
- Logarithm laws: $\log(mn) = \log m + \log n$; $\log(m/n) = \log m - \log n$; $\log m^k = k \log m$; $\log_a a = 1$; $\log_a 1 = 0$; change of base $\log_a b = \log_c b / \log_c a$.
- Constants can also be written as logarithms: $4 = 4 \ln e = \ln e^4$; $2 = \log_{10} 100$; $9 = \log_{10} 10^9$. This is a key step in 2025F-Q7(a), 2020F-Q6(c), 2024A-Q15(c), 2019A-Q17(b)(iii).
- Calculus: $(\ln f)' = f'/f$; $\int f'/f \, dx = \ln|f| + c$ (in exams generally $f(x) > 0$, the key often writes $\ln f(x) + c$); $(e^{kx})' = ke^{kx}$.

Marker reminders

- Solving $2 \ln x = \ldots$: $\ln x$ requires $x > 0$, the negative root must be rejected (the 2025F-Q7 key explicitly requires only one solution).
- "How many times" questions on logarithmic scales: the difference in readings is not the answer; the difference must still be converted to a power of 10 (the 2016A-Q12 key has 1 mark for "subtracts Richter magnitudes", then converting to exponential form). The 2023 report calls for stronger interpretation and reading of logarithmic scales (see 2023F-Q4(c): reading values from the graph of $y = \log_{10} x$).
- When a question requires using the graph, show the reading process on the graph (2024A-Q15(a): You must show clearly how you have used the graph).
- Application questions: include units. If asked for rate of change, give a signed value (e.g. the answer to 2018A-Q9(c) is -0.164 mg/L/h); only if asked for rate of decrease do you write it as a positive number and state decreasing (2024A-Q8(c) key: "converts to a rate of decrease").

Pattern 1: Express logarithms in terms of given letters (CF)

Recognise: "Let $p = \ln 2$, $q = \ln 3$, $r = \ln 5$. Express $\ln 6.25$ in terms of …" "$\log_{10} 2 = x$, $\log_{10} 7 = y$".

Approach: factorise the argument into products, quotients and powers of the given bases: $6.25 = 25/4 = 5^2/2^2 \Rightarrow 2r - 2p$; $17.5 = 7 \times 10 / 4 \Rightarrow y + 1 - 2x$; $14 = 2 \times 7$. Change-of-base type (2025F-Q7(b)): $m = \log_3 6$, $n = \log_6 5$, find $\log_3 10$. First $\log_3 10 = \log_3 2 + \log_3 5$; then $\log_3 2 = \log_3 6 - 1 = m - 1$, $\log_3 5 = \log_3 6 \cdot \log_6 5 = mn$ $\Rightarrow \log_3 10 = m - 1 + mn$. e^{p+q} type: e^{ln2 + ln3} = e^{ln 6} = 6 (index laws plus inverse relationship).

Marking points: ✓ factorise the argument ✓ use the correct logarithm laws ✓ final expression.

Past questions: 2017F-Q7, 2023F-Q2(a)(b), 2024F-Q5(a)(b), 2025F-Q7(b).

Pattern 2: Solve exponential / logarithmic equations (CF exact values)

Approach

- Collect like terms: $4e^{2x} = 81 - 5e^{2x} \Rightarrow 9e^{2x} = 81 \Rightarrow e^{2x} = 9 \Rightarrow x = \ln 3$.
- Quadratic in e^x: let $u = e^x$, factorise, and reject roots with $u \le 0$ (2020F-Q7(a): $e^x(e^x - 4) = 0 \Rightarrow$ only $x =$

ln 4).

- Logarithmic equations: first combine all terms into a single logarithm ($\log_2(x+y) + 2 = \log_2(x - 2y) \Rightarrow \log_2[4(x+y)] = \log_2(x - 2y)$), then remove the logarithms, and finally check the domain.

Past questions: 2017F-Q3, 2016F-Q1(b), 2020F-Q7(a), 2024F-Q5(c), 2025F-Q7(a).

Pattern 3: Logarithmic function graphs and transformations

Recognise: "sketch $y = \ln(x - 2) + 1$" "determine the equation of g" "explain why p = 5" "find m, q".

Approach

1. $y = m \log_a(x - p) + q$: vertical asymptote $x = p$; always passes through $(p + 1, q)$; increasing for $m > 0$ when $a > 1$.
2. Three essential features when sketching (1 mark each in the key): asymptote (draw and label its equation, conventionally dashed), one key point (e.g. (3, 1)), correct shape (including behaviour near the asymptote).
3. Determine parameters from the graph: first use the asymptote to find p, then substitute two points to solve for m and q (2021A-Q15).
4. Recognising transformations: $\ln(4x) = \ln x + \ln 4 \rightarrow$ vertical translation up by $\ln 4$; $\ln\sqrt{x} = \frac{1}{2}\ln x \rightarrow$ vertical dilation by ½; $\ln x + 4 = \ln(e^4 x) \rightarrow$ horizontal dilation, factor $1/e^4$ (2020F-Q6).
5. $\log_2(1/(x-1)) = -\log_2(x - 1)$: first reflect in the x-axis, then shift right by 1 (2022F-Q4(c)).

Using a graph to solve equations or estimate: e.g. $\sqrt{7} \Rightarrow \log_2\sqrt{7} = \frac{1}{2}\log_2 7 \approx \frac{1}{2} \times 2.8 = 1.4 \Rightarrow$ read $x \approx 2.6$ (2022F-Q4(a)(ii)).

Past questions: 2018A-Q8, 2019F-Q4, 2020F-Q6, 2021A-Q15, 2022F-Q4, 2024F-Q5(d).

Pattern 4: Exponential growth/decay models ★ CA frequently tested

Recognise: $P = P_0 e^{kt}$, C = 4e^{−0.05t}, $T = T_0 - 175e^{-0.07t}$, half-life, "in danger if below …".

Standard process

1. Initial value: substitute t = 0 (with units).
2. Find k: use the half-life ½ = e^{k·100.5}, or use two data points. Round to the required number of significant figures.
3. Time to reach a value: set the model equal to the target value and solve using ln; convert to a specific time according to the context, and round as required by the question. Example: in 2018A-Q9(d), 4e^{−0.05t} = 2.5 ⇒ t = ln 1.6 / 0.05 ≈ 9.40 h ⇒ 9 am + 9 h 24 min = 6:24 pm (for "latest", round down; the key notes 6:25 is too late). Roots outside the domain must be explicitly rejected (2025A-Q9(b)).
4. Rate of change: dP/dt = kP, substitute the time; for signs see the reminder above.
5. Piecewise models: the initial value of the later model = the value at the end of the previous section (2017A-Q16(c), 2023A-Q6(e)).
6. Long-term behaviour: as $t \rightarrow \infty$, e^{−kt} → 0, so $T \rightarrow T_0$ and the rate of change → 0 (2020A-Q15(e) three points: rate of change → 0, temperature approaches a constant, the constant equals T_0).

Marking points: ✓ set up equation ✓ solve ✓ round or convert time as required ✓ differentiate ✓ substitute value.

Past questions: 2016A-Q9, 2016S-A10, 2017A-Q16, 2018A-Q9, 2020A-Q15, 2022A-Q14, 2023A-Q6, 2024A-Q8, 2025A-Q9(b).

Pattern 5: Logarithmic scales (decibels, Richter, site ranking, moment magnitude)

Recognise: $L = 10 \log(I/I_0)$, $M = \log_{10}(A/A_0)$, $R = 2 \log_{10}(S/S_0)$, asked "how many times larger / more intense".

Approach (ratio questions)

1. General form $L = k \cdot \log_{10}(I/I_0)$. Convert the two readings to exponential form separately: $I_1 = I_0 \cdot 10^{L_1/k}$, $I_2 = I_0 \cdot 10^{L_2/k}$.
2. Divide, and I_0 cancels: $I_2/I_1 = 10^{(L_2-L_1)/k}$. k varies by question: decibels L = 10 log ⇒ 90 dB versus 60 dB is $10^3 = 1000$ times; sound pressure D = 20 log (2016S-A13) ⇒ a 30 dB difference is $10^{1.5} \approx 31.6$ times; Richter k = 1 (2016A-Q12) ⇒ $10^{2.1} \approx 126$ times; site ranking R = 2 log (2023F-Q4).
3. Find the reference value I_0 or S_0: substitute a set of (L, I) and convert to exponential form to solve.

Linearisation (log against t is a straight line): $\log_{10}P = At + B \Rightarrow P = 10^B \cdot (10^A)^t$; the gradient A reflects how fast the growth is (2019A-Q17(c) comparing growth rates at two temperatures requires referring to the gradient or the base).

Marking points: ✓ convert to exponential form ✓ write the ratio and simplify ✓ conclusion ("… times").

Past questions: 2016A-Q12, 2016S-A13, 2018A-Q18, 2019A-Q17, 2022A-Q14(c)–(e), 2023F-Q4, 2024A-Q15, 2025A-Q17.

Pattern 6: Logarithmic models + differentiation + drawing conclusions

Identify: $s = 40.6 \ln(0.15D)$, $A(t) = b \log_4(t+1) + c$, $P(x) = 20 \ln(x+a)/(x+5)$.

Approach

- Determine the parameters from the given points: $A(0) = 21 \Rightarrow c = 21$; $A(1) = 53 \Rightarrow b \cdot \log_4 2 = 32 \Rightarrow b = 64$.
- Interpretation from the derivative: $ds/dD = 40.6/D$ decreases as D increases ⇒ for the same 8 m decrease, the speed limit drops by less on roads with a larger original stopping distance (2025A-Q11(c)).
- Decide whether the target can be achieved: solve $A(t) = 493$ and check whether t is reasonable (2024A-Q17(c)); draw a conclusion with a reason.
- Round according to the context: round quantities down (2024A-Q17(b)); compare adjacent integers (2020A-Q13 compares 74 and 75).
- "during which week": x is the number of weeks from the start, so $1 < x < 2$ falls in the 2nd week, not the 1st week (the key for 2021A-Q16(d) writes "occurs in the second week").

Past questions: 2018F-Q6, 2020A-Q13, 2021A-Q16, 2024A-Q17, 2025A-Q11.

Pattern 7: Calculus of e^x / ln x (cross-reference)

The methods for these questions are covered in the differentiation and integration handouts; only the questions are listed here, so you can drill the "exponential and logarithmic" topic:

- Use d/dx(x ln x) to find ∫ln x: 2018F-Q7, 2023F-Q2(c) → see `02_Integration_Methods.md` Pattern 2.
- Areas involving e^x and ln x (including areas of inverse functions), and the correspondence between $\int 1/x$ and ln areas: 2017F-Q5, 2019F-Q5, 2020A-Q11, 2021F-Q7 → Integration handout Patterns 5, 6.
- Find f from f′, and $\int f'/f$: 2020A-Q10 → Integration handout Pattern 1.
- Use the increment formula to estimate ln 2.02: 2021F-Q3 → Differentiation handout Pattern 7.
- Find stationary points and sketch graphs of functions involving ln and e^x: 2016A-Q13, 2017A-Q14, 2020F-Q7(b)–(d), 2019A-Q12 → Differentiation handout Pattern 4.
- Derivative of a^x and the definition of e: 2018A-Q14.